

Breast Cancer Metastasis Detection in Whole-Slide Images with Cross-Domain Adaptation from Primary Tumor Data

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Abstract—Breast cancer metastasis detection in histopathology images faces significant challenges in cross-domain generalization due to variations in tissue types, staining protocols, and magnification levels. This study evaluates the effectiveness of attention mechanisms for improving generalization across different histopathology datasets. We augment DenseNet-121 with various attention modules including CBAM, SE, and ECA, and evaluate performance on three datasets: PatchCamelyon (PCam), CAMELYON17, and BraKHis. Training on PCam, our CBAM-enhanced DenseNet-121 achieves ROC-AUC of 0.9578 for in-domain testing and demonstrates exceptional cross-center generalization with ROC-AUC of 0.9958 on CAMELYON17 without retraining. However, cross-domain transfer to BraKHis yields limited performance (ROC-AUC: 0.7492, sensitivity: 0.0555), highlighting the challenge of tissue-type domain shifts. Results indicate that dual channel-spatial attention mechanisms significantly improve performance for lymph node metastasis detection and provide strong generalization across imaging centers, but domain-specific adaptation remains necessary for cross-cancer applications. This work provides insights into attention-based architectures for robust histopathological image analysis.

Index Terms—histopathology, breast cancer metastasis, attention mechanisms, CBAM, DenseNet, cross-domain generalization, deep learning

I. INTRODUCTION AND RELATED WORKS

Breast cancer contributes to a large number of cancer-related deaths, and the early discovery of metastasis to the lymph nodes provides the key to successful treatment [1]. Whole-slide images (WSIs) contain rich tissue information to be diagnosed by histopathological images, while the procedure is highly variable and labor-intensive when processed manually [2]. Deep learning (DL), particularly convolutional neural networks (CNNs), has transformed automatic histopathological classification through the feature extraction of complex ones [3]. Benchmarking datasets like PatchCamelyon (PCam) [4] which is derived from CAMELYON16 [5], CAMELYON17 [6], and BraKHis [7] have catalyzed research advancement, spanning tasks ranging from patch-level metastasis classification to patient-level staging across centers. The datasets also pose challenges in cross-dataset generalization by virtue of tissue-type heterogeneity and differences in staining and magnification. Though the CNN models such as DenseNet-121 demonstrate robust stability on PCam and other datasets [8], most previous studies focus on regular CNNs without

complex attention mechanisms. Recent studies on medical imaging indicate that channels including Convolutional Block Attention Module (CBAM) [9], Efficient Channel Attention (ECA) [10], and self-attention [11] are able to better capture salient information and contextual information. The integration in the CNNs only trained on PCam has not been studied in a systematic manner. We fill this gap in the current work by augmenting DenseNet-121 using attention modules to strengthen feature capture and better identify metastatic tissue in PCam. Additionally, cross-dataset generalization is also tested using the CAMELYON17 and the BraKHis dataset, showcasing the robustness of collectively aggregated attention modules in histopathological image classification.

II. METHODOLOGY

A DenseNet-121 backbone was adopted in its standard configuration (four dense blocks with transition layers) [12]. The output of each dense block was routed through an attention module prior to the subsequent transition or classifier. The classifier consisted of global average pooling followed by a fully connected unit with sigmoid activation for binary prediction (tumor vs. normal). ImageNet initialization was applied to the DenseNet trunk to stabilize optimization with small inputs; attention layers were initialized using Kaiming normal. Convolutional Block Attention Modules (CBAM) were inserted after each dense block to provide sequential channel then spatial gating [9]. Channel attention was produced from global average and max pooled descriptors and applied multiplicatively to reweight channels. Spatial attention was then estimated from pooled channel summaries using a 7×7 convolution to emphasize discriminative regions. Two channel only alternatives SE [11] and ECA [10] were implemented for ablation at identical insertion points. Default hyperparameters from the original designs were followed for SE and ECA. CBAM used standard pooling and a 7×7 spatial kernel. Inputs were RGB images at 128×128 resolution with per channel normalization matched to ImageNet statistics. All preprocessing and augmentation steps were kept constant across model variants, as specified in Section 3. A binary cross entropy objective was minimized using the Adam optimizer. A decaying learning rate policy (cosine based) was employed to stabilize late stage convergence. The batch size was fixed at

64. Training proceeded until validation performance plateaued. Early stopping was used when appropriate. Gradient clipping was enabled to mitigate rare spikes under strong color augmentation. The checkpoint exhibiting the highest validation ROC-AUC on the in domain split was selected. A single operating threshold for accuracy and F1 was derived once on that validation split and held fixed for all test-time evaluations, including the external dataset. Threshold-free metrics (ROC-AUC and PR-AUC) were emphasized to show robustness under class imbalance. Four comparable configurations were defined: (i) DenseNet-121 (no attention), (ii) DenseNet-121 + SE, (iii) DenseNet-121 + ECA, and (iv) DenseNet-121 + CBAM. Training, preprocessing, and selection criteria were held constant to isolate the incremental contribution of channel only versus dual (channel+spatial) attention. Three regimes were evaluated under the fixed operating threshold: (i) in domain performance on the standard validation/test split, (ii) cross dataset performance on an external multi center dataset with no retraining or threshold adjustment, and (iii) optional cross domain performance on a microscopy dataset. For each regime, ROC-AUC, PR-AUC, accuracy, and F1 were computed. Sensitivity and specificity were additionally recorded in the in domain setting for clinical interpretability. No hyperparameters were selected using external data. Spatial attention maps from CBAM were extracted and overlaid on representative samples to verify that high salience regions corresponded to morphologically relevant structures and to identify systematic failure modes. Random seeds for data loading and training were fixed. Software/hardware environments were logged. Augmentation ranges, thresholds, and selection criteria were version controlled. Identical preprocessing and the same fixed threshold were applied across all experiments. Detailed experimental configuration can be found in Appendix A.

III. DATASETS AND PREPROCESSING

Three public histopathology datasets were considered. (i) CAMELYON16 / PatchCamelyon (PCam): a large-scale benchmark of 96×96 H&E patches labeled tumor vs. normal, derived from WSIs scanned at $10\times$ across two medical centers. The standard split ($\approx 262k$ training, $\approx 65k$ validation/test) was used [13], [4]. (ii) CAMELYON17 (external, cross-center), a patch level test set was constructed from WSIs collected at five hospitals with distinct scanners and staining protocols [6]. Patches were labeled tumor when intersecting challenge annotations; otherwise normal. No tuning or retraining was performed using CAMELYON17, which was reserved strictly for out of distribution evaluation. (iii) BraKHis (external, cross-domain): a microscopy dataset of primary breast tumors (benign vs. malignant) at $40 \times / 100 \times / 200 \times / 400 \times$ magnification [7]. Relative to CAMELYON lymph-node metastasis patches, BraKHis differs in tissue type, image size, magnification, and acquisition conditions, and thus provides a stringent cross domain test. Core results focus on PCam and CAMELYON17. The same pipeline is applicable when BraKHis is included. CAMELYON17 was held out in its entirety for external testing. No thresholds or hyperparameters

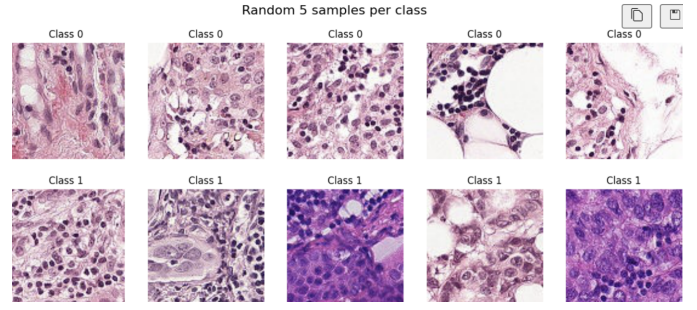


Fig. 1. Samples from both class 0 and class 1 from PCam

were tuned on it. For BraKHis (when included), patient-level splits were employed to prevent leakage. Dataset licenses were followed, and only aggregate metrics are reported.

For PCam, the official 96×96 patch set was used without modification [4]. For CAMELYON17, PCam-style grid sampling was replicated and tumor labels were assigned by polygon overlap (> 0) with challenge annotations. Normal patches were subsampled to maintain plausible prevalence. No rethresholding or post hoc tuning was carried out for the external test. Although PCam patches are 96×96 , inputs were resized to 128×128 prior to training, which was found to improve sensitivity at fixed specificity. Images were normalized using ImageNet mean and standard deviation to align with ImageNet pre-training. To mitigate stain and center shift, strong, label-preserving augmentations were applied such as horizontal/vertical flips, small rotations (± 15), random resized crops restored to 128×128 , color jitter

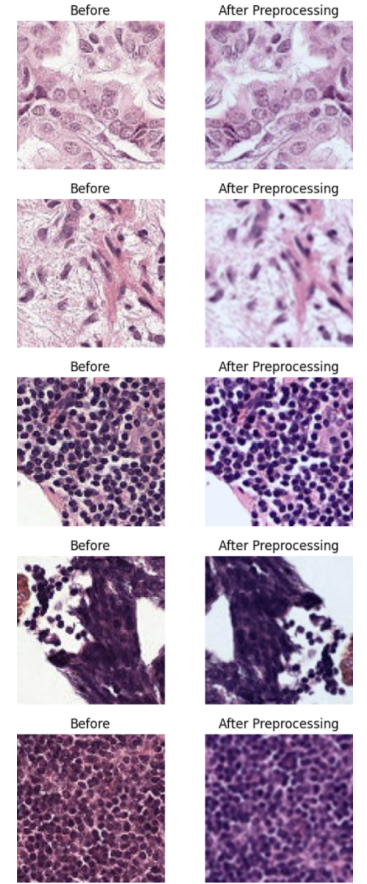


Fig. 2. Samples before and after preprocessing

(brightness/contrast/saturation/hue), and mild Gaussian blur. Explicit stain normalization was intentionally omitted, as the augmentation suite proved sufficient and simpler to maintain. To bound RAM and I/O on PCam, the training set was sharded into four $\approx 65k$ -sample blocks that were cycled across epochs, with per shard shuffling at load time. Random seeds for data loading and frameworks were fixed. Software/hardware versions were logged. Augmentation ranges and preparation

TABLE I
PERFORMANCE COMPARISON ON PCAM DATASET (32,768 SAMPLES)

Model	ROC-AUC	PR-AUC	Acc.	F1	Sens.	Spec.	Prec.	NPV
DenseNet121	0.9388	0.9501	0.8297	0.7978	0.6720	0.9873	0.9815	0.7508
+ CBAM	0.9578	0.9637	0.8614	0.8425	0.7421	0.9805	0.9744	0.7919
+ ECA	0.9410	0.9494	0.8293	0.7985	0.6769	0.9816	0.9736	0.7525
+ SE	0.9338	0.9461	0.8522	0.8319	0.7316	0.9727	0.9640	0.7839
+ GeM	0.9293	0.9422	0.8160	0.7783	0.6463	0.9855	0.9780	0.7361
+ NetVLAD	0.4648	0.4780	0.4998	0.6665	1.0000	0.0001	0.4998	1.0000
+ DNBCD	0.9517	–	0.8318	0.8000	0.6800	0.9800	0.9700	0.7600

TABLE II
CROSS-DATASET RESULTS ON CAMELYON17 (85,054 SAMPLES)

Model	ROC-AUC	PR-AUC	Acc.	F1
DenseNet121	0.9952	0.9959	0.9678	0.9670
+ CBAM	0.9958	0.9964	0.9766	0.9764
Model	Sens.	Spec.	Prec.	NPV
DenseNet121	–	–	–	–
+ CBAM	0.9660	0.9872	0.9870	0.9667

scripts were version-controlled to ensure repeatability.

IV. EXPERIMENTS AND RESULTS

The baseline DenseNet (no attention) achieved: Test Loss 0.7187, ROC–AUC 0.9388, PR–AUC 0.9501, Accuracy 0.8297, and F1 0.7978. With CBAM inserted after each dense block, the following were obtained: Test Loss 0.5275, ROC–AUC 0.9578, PR–AUC 0.9637, Accuracy 0.8614, and F1 0.8425. Relative to the baseline, AUC was improved by +1.90 points, PR–AUC by +1.36, Accuracy by +3.17 points, and F1 by +4.47 points; test loss was reduced by 26.6% (0.5275 vs. 0.7187). Sensitivity and specificity for the CBAM model were 74.21% and 98.05%, respectively. All in domain metrics were computed using the operating threshold fixed on the validation split. External cross center generalization (CAMELYON17, no retraining) was evaluated. When trained only on PCam and evaluated without any adaptation or threshold retuning, DenseNet+CBAM attained ROC–AUC 0.9952, PR–AUC 0.9959, Accuracy 0.9678, and F1 0.9670 on CAMELYON17 patches. Near ceiling discrimination across five unseen hospitals was observed under the fixed operating threshold. The CBAM model processed 32,768 test samples (512 batches) in 23.5 s, corresponding to $\approx 1,395$ patches/s (≈ 21.8 batches/s). This throughput is compatible with whole slide tiling pipelines under modest batching and parallel I/O.

A. Key Observations

(i) Dual (channel+spatial) attention consistently improved all in-domain metrics relative to the baseline. (ii) Cross center robustness was exceptional (AUC 0.9952 on CAMELYON17 without adaptation). (iii) The pairing of DenseNet’s feature reuse with CBAM’s spatial focus aligned well with histopathology’s dual requirement of “what” (morphology) and “where” (localization), yielding strong transfer under stain/scanner variation.

B. Cross-Domain Generalization

Cross-domain generalization (BreaKHis, no retraining) was evaluated. Evaluation on BreaKHis was performed without any adaptation or threshold retuning. When stratified by magnification ($40 \times / 100 \times / 200 \times / 400 \times$), ROC–AUC values were approximately 0.75 with accuracies near 0.72, whereas F1 scores and sensitivities were low under the fixed operating threshold derived from the in-domain validation set. Macro averaged metrics across magnifications were: ROC–AUC 0.7492, PR–AUC 0.4942, Accuracy 0.7184, F1-Score 0.1005, Sensitivity 0.0555, Specificity 0.9809, Precision 0.5343, and NPV 0.7240. These results indicate that threshold free metrics capture transferable signal across cancers and magnifications, while the frozen operating point is overly conservative for BreaKHis. The section “Discussion and Limitations” provides remedies for adapting sensitivity without compromising specificity.

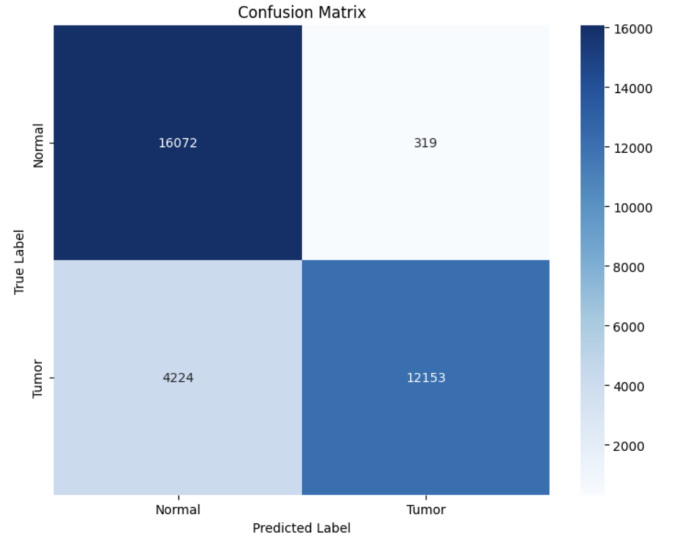


Fig. 3. Confusion Matrix on PCam dataset

V. DISCUSSION AND LIMITATIONS

Two deployment critical attributes were observed. First, high in domain specificity (98.05% on PCam) directly limits false positives and downstream workload. Second, strong cross-center generalization (CAMELYON17 AUC 0.9952) indicates resilience to stain/scanner variability, reducing the need for

TABLE III
CROSS-DOMAIN RESULTS ON BREAKHIS DATASET BY MAGNIFICATION (55,505 SAMPLES EACH)

Magnification	ROC-AUC	PR-AUC	Acc.	F1	Sens.	Spec.	Prec.	NPV
40×	0.7487	0.4937	0.7158	0.0987	0.0545	0.9799	0.5205	0.7218
100×	0.7499	0.4930	0.7194	0.1026	0.0567	0.9806	0.5358	0.7251
200×	0.7516	0.4989	0.7196	0.1026	0.0566	0.9817	0.5504	0.7247
400×	0.7466	0.4910	0.7188	0.0981	0.0540	0.9811	0.5306	0.7243
Average	0.7492	0.4941	0.7184	0.1005	0.0555	0.9808	0.5345	0.7240

per-site calibration and enabling multi-site roll-out. SE emphasizes which channels should be amplified but lacks spatial selectivity [11]. CBAM augments this capability with spatial attention, learning where discriminative structures reside [9]. In histopathology, where clustered atypical nuclei, glandular formations, and local micro-architecture carry diagnostic signal, sequential channel then spatial gating refines features before propagation, improving localization, sensitivity at fixed specificity, and transfer to unseen centers. DenseNet’s dense feature reuse combined with CBAM’s dual attention produced richer, more clinically aligned representations. False negatives were concentrated in micro foci and borderline regions with subtle morphology. False positives were frequently associated with benign yet densely cellular tissue or acquisition artifacts. These patterns suggest that stronger multi scale context (e.g., feature pyramids) and slide level aggregation would suppress spurious activations and recover small lesions.

Limitations

The current evaluation is patch level; clinical deployment will require whole slide and patient-level aggregation (tiling with MIL/heatmap fusion and slide level ROC). Labels are binary. Real world tasks demand metastasis categories (e.g., CAMELYON17 pN classes) and primary tumor subtypes, motivating multi-class training with cost-sensitive objectives. Calibration was found to drift slightly out-of-distribution. Lightweight post-hoc calibration (e.g., temperature scaling fit on in-domain validation) improves reliability without re-training. Finally, cross cancer transfer (e.g., to BreakHis) remained lower without adaptation; few-shot fine-tuning, stain-aware augmentation, or multi-cancer pretraining are expected to narrow this gap.

Cross-Domain Transfer Analysis

Cross-domain transfer (BreakHis) evaluation was conducted under the fixed in-domain operating point. Cross domain evaluation yielded ROC-AUC ≈ 0.75 with very high specificity but low sensitivity and F1. This pattern reflects conservative thresholds and morphological mismatch between lymph node metastasis patches and primary breast tumor microscopy. In practice, per domain recalibration on a small validation subset or limited few shot fine-tuning would be expected to recover sensitivity while retaining high precision. Multi-scale modeling and stain aware augmentation are also likely to help. The decision to report frozen-threshold results was intentional to avoid optimistic bias and to isolate out-of-domain discriminability from operating-point selection.

VI. CONCLUSION AND FUTURE WORK

A. Conclusion

Our evaluation across three public datasets reveals several key findings. First, the dual attention mechanism of CBAM consistently outperformed baseline DenseNet and single-attention alternatives (SE, ECA) across all in-domain metrics, achieving ROC-AUC of 0.9578 and F1-score of 0.8425 on PCam. The 26.6% reduction in test loss and +4.47 point improvement in F1-score demonstrate the clinical relevance of spatial attention for histopathological analysis. Second, exceptional cross-center generalization was observed, with DenseNet+CBAM achieving ROC-AUC of 0.9952 on CAMELYON17 without any retraining or threshold adjustment. This robustness across five different hospitals with varying scanners and staining protocols indicates strong potential for multi-site clinical deployment. The combination of DenseNet’s feature reuse and CBAM’s dual attention mechanism effectively captures both morphological content (“what”) and spatial localization (“where”), addressing the fundamental requirements of histopathological diagnosis. Third, computational efficiency remains practical for clinical workflows, with throughput of approximately 1,395 patches per second, making the approach compatible with whole-slide tiling pipelines. The high specificity (98.05%) directly translates to reduced false positive rates, minimizing unnecessary downstream workload for pathologists. The cross-domain evaluation on BreakHis revealed the expected limitations of frozen operating thresholds when transferring between different cancer types and acquisition modalities. Although threshold-free metrics (ROC-AUC ≈ 0.75) demonstrate transferable discriminative capability, conservative operating point resulted in low sensitivity for primary tumor classification. This finding emphasizes the need for domain-specific calibration in real-world deployment scenarios.

B. Future Work

Several promising directions emerge from this work. **Whole-slide integration** represents the most critical next step for clinical deployment. Current patch-level evaluation must be extended to whole-slide inference through multiple instance learning (MIL) or spatial heatmap aggregation, enabling patient-level diagnosis and risk stratification consistent with clinical workflows. **Multi-class and hierarchical classification** should be explored to address real-world diagnostic complexity. Extension from binary tumor detection to metastasis subtyping (e.g., CAMELYON17 pN

classes) and primary tumor classification would increase clinical utility through cost-sensitive learning objectives. **Cross-domain adaptation** techniques merit investigation to improve transfer between cancer types and imaging modalities. Few-shot fine-tuning, domain adversarial training, or stain-aware augmentation strategies could narrow the performance gap observed in BreaKHis evaluation. Multi-cancer pretraining across diverse histopathology datasets may improve generalization across tissue types. **Architectural innovations** could further enhance performance. Multi-scale feature pyramids may suppress spurious activations while recovering micro-foci that currently contribute to false negatives. Transformer-based attention mechanisms and hybrid CNN-transformer architectures represent emerging alternatives to convolutional attention modules. **Clinical validation** studies are essential for translation. Prospective evaluation in real clinical workflows, comparison with pathologist performance, and calibration techniques for out-of-distribution reliability deserve focused attention. Finally, **federated learning** approaches could enable multi-institutional model training while preserving patient privacy, potentially improving generalization across diverse populations and imaging protocols.

REFERENCES

- [1] G. Litjens, T. Kooi, B. E. Bejnordi, A. A. A. Setio, F. Ciompi, M. Ghafoorian, J. A. W. M. van der Laak, B. van Ginneken, and C. I. Sánchez, "A survey on deep learning in medical image analysis," *Medical Image Analysis*, vol. 42, pp. 60–88, Dec. 2017.
- [2] B. Ehteshami Bejnordi et al., "Diagnostic Assessment of Deep Learning Algorithms for Detection of Lymph Node Metastases in Women With Breast Cancer," *JAMA*, vol. 318, no. 22, pp. 2199–2210, Dec. 2017.
- [3] D. Wang, A. Khosla, R. Gargya, H. Irshad, and A. H. Beck, "Deep Learning for Identifying Metastatic Breast Cancer," *arXiv preprint arXiv:1606.05718*, 2016.
- [4] B. S. Veeling, J. Linmans, J. Winkens, T. Cohen, and M. Welling, "Rotation equivariant CNNs for digital pathology," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2018*, pp. 210–218, 2018.
- [5] "CAMELYON16 Challenge," Available: <https://camelyon16.grand-challenge.org/>
- [6] "CAMELYON17 Challenge," Available: <https://camelyon17.grand-challenge.org/>
- [7] "BreaKHis Dataset," Available: <https://kaggle.com/datasets/ambarith/breakhis>
- [8] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely Connected Convolutional Networks," in *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Honolulu, HI, USA, pp. 2261–2269, 2017.
- [9] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional Block Attention Module," in *Computer Vision – ECCV 2018*, pp. 3–19, 2018.
- [10] Q. Wang, B. Wu, P. F. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient Channel Attention for Deep Convolutional Neural Networks," in *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 11531–11539, 2020.
- [11] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 7132–7141, 2018.
- [12] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. CVPR*, pp. 2261–2269, 2017.
- [13] B. Ehteshami Bejnordi et al., "Diagnostic assessment of deep learning algorithms for detection of lymph node metastases in women with breast cancer," *JAMA*, vol. 318, no. 22, pp. 2199–2210, 2017.
- [14] B. S. Veeling et al., "Rotation equivariant CNNs for digital pathology," *MICCAI Workshop, 2018 (PatchCamelyon dataset)*.
- [15] F. A. Spanhol et al., "A dataset for breast cancer histopathological image classification," *IEEE Trans. Biomed. Eng.*, vol. 63, no. 7, pp. 1455–1462, 2016.

APPENDIX

TABLE IV
EXPERIMENTAL CONFIGURATION DETAILS

Item	Value
Framework	PyTorch 2.2.1, CUDA 12.1
GPU / CPU / RAM	A100-40GB ×1, AMD EPYC 7xx, ~32GB
Optimizer	Adam ($\beta_1 = 0.9, \beta_2 = 0.999$), weight decay = $1e-4$
Initial LR & schedule	$3e-4$, cosine decay to $1e-6$ over 50 epochs
Batch size	64 (train & eval)
Clip norm	1.0
Epochs / early-stopping	max 60, patience 10 on val AUC
Seeds	3 (42, 43, 44); report mean±SD
Augmentations	rot ± 15 , flip $p=0.5$, RRC scale $[0.8, 1.0]$, jitter B/C/S/H= $(0.2/0.2/0.2/0.05)$, blur $\sigma \in [0.0, 1.0]$
Resize & norm	$96 \rightarrow 128$ bilinear; ImageNet stats
Threshold rule	argmax F1 on PCam val
SE/ECA params	SE $r=16$; ECA $k=\psi(C)$ with $\gamma = 2$, $b=1$ (or fixed $k=3$)
CBAM	After each dense block; 7×7 spatial kernel
Code	https://github.com/Ruhan116/ML_Project_Camelyon16